COUNTING FIGURES

Squares, Rectangles and Triangles in a Figure

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 21

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: a “how many squares are there in the figure” question appears in English 2022 (Q32).

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). One counting-figures question appears (English 2022 Q32). It is solved with its original figure. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The one PYQ and what it asks
2	Foundation	Counting formulas and the systematic method
3	Solved PYQs	English 2022 Q32 with its original figure
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

One question, in English 2022

Paper	Q.	Question	Figure	Ans
English 2022	32	How many squares are there in the figure?	4 × 4 grid with diagonals	B (35)
English 2019 / Mathematics 2024 / Science 2024 / Social Studies 2023	–	none		

PYQ INSIGHT: Only one question in five papers, but counting figures is a standard reasoning topic. The PYQ is a **4 × 4 grid with diagonals**: counting only the squares along the grid lines gives 30, and the diagonals add the tilted squares. Always count by **size**, then add the tilted ones.

Pattern	PYQs	Priority
Squares in a grid, including tilted squares formed by diagonals	1	★★★ Must
Rectangles in a grid	0	★★ Expected
Triangles in a figure with lines from the apex	0	★★ Expected
Triangles in a figure with parallel lines or a square with diagonals	0	★★ Expected

CHAPTER 2

Foundation

Formulas and a systematic method

2.1 Formulas

Figure	Count	Example
Squares in an n × n grid	$n^2 + (n-1)^2 + \ldots + 1 = n(n+1)(2n+1)/6$	4 × 4: 16 + 9 + 4 + 1 = 30
Squares in an m × n grid (m ≤ n)	Sum of (m−k+1)(n−k+1) for k = 1 … m	3 × 4: 12 + 6 + 2 = 20
Rectangles in an m × n grid (squares included)	m(m+1)/2 × n(n+1)/2	3 × 4: 6 × 10 = 60
Triangles with p lines from the apex to the base	(p + 1)(p + 2)/2	p = 4: 15
Same, plus q lines parallel to the base	(p + 1)(p + 2)/2 × (q + 1)	p = 2, q = 2: 6 × 3 = 18

2.2 Method

Step	Do this
1	Label the smallest regions (or the lines) with letters or numbers.
2	Count by size: 1 unit, 2 units, 3 units … so that no square is missed or counted twice.
3	Add the tilted squares made by the diagonals (their sides lie on the diagonals).
4	Check the total with the formula, if a plain grid is involved.

NOTE: The diagonals of the cells of a grid create tilted squares only where the four diagonals around a point form a diamond. In English 2022 Q32 the diagonals alternate between cells, and they form 5 tilted squares (centred at five interior grid points). So 30 + 5 = 35.

SHORTCUT: For a grid, count squares by size and use the formula; for triangles, count the triangles that have the **same apex** and multiply by the number of levels.

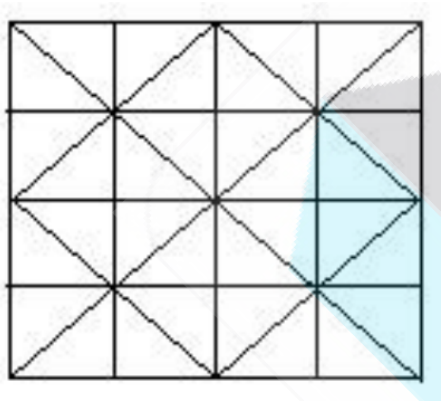
CHAPTER 3

Solved PYQs

English 2022 Q32 with its original figure

English 2022 Q32

How many squares are there in the following figure? (A) 30 (B) 35 (C) 28 (D) 33



1. The figure is a 4 × 4 grid. Squares along the grid lines: 16 of size 1, 9 of size 2, 4 of size 3, 1 of size 4: total 30.
2. The diagonals alternate between cells, so around five interior points they form tilted squares (diamonds).
3. Total = 30 + 5 = 35.
4. The scan is small, so the count of tilted squares is taken from the pattern of the diagonals; the official key is 35.
Answer: Option B (35)

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

• **Grid:** use n(n+1)(2n+1)/6 for squares and m(m+1)n(n+1)/4 for rectangles.

• **Triangles from an apex:** count the segments on the base; triangles = segments chosen in pairs (including the whole base).

• **Tilted squares:** look for diamonds formed by four diagonals meeting around a point.

• **Never count twice:** tick or letter each square as you count it.

• **Time:** 45 seconds per question.

Trap	How to avoid it
Forgetting larger squares	Count sizes 1, 2, 3 … up to the full grid.
Missing tilted squares	Diagonals make diamonds; count them separately.
Counting rectangles as squares	A rectangle that is not a square does not count in a “squares” question.
Double counting	Count by size, not by location.

Scope

Related PYQs / topics	Where
Embedded figures (finding a figure inside another)	Embedded Figures (Topic 22)
Triangle series in the syllabus line	Triangle (earlier topic)

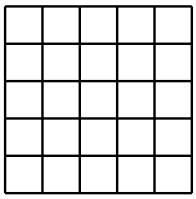
CHAPTER 5

Practice Set

6 original questions; answers on the next page

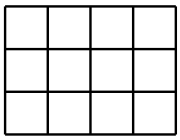
A. Squares and rectangles

Q1. How many squares are there in a 5 × 5 grid?



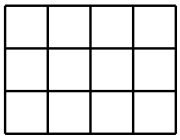
A. 50 B. 55 C. 45 D. 60

Q2. How many rectangles (squares included) are there in a 3 × 4 grid?



A. 40 B. 30 C. 50 D. 60

Q3. How many squares are there in a 3 × 4 grid (3 rows, 4 columns)?



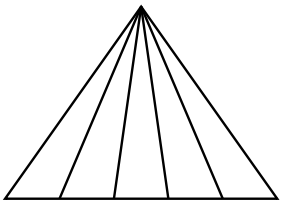
A. 20 B. 12 C. 18 D. 24

Q4. How many squares of size larger than 1 × 1 are there in a 4 × 4 grid?

A. 16 B. 30 C. 14 D. 9

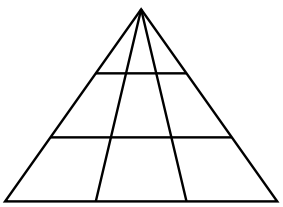
B. Triangles

Q5. A triangle has 4 lines drawn from the apex to the base. How many triangles are there?



A. 10 B. 5 C. 15 D. 20

Q6. A triangle has 2 lines from the apex to the base and 2 lines parallel to the base. How many triangles are there?



A. 9 B. 18 C. 12 D. 15

Answer key

Q	Answer	Reason
1	B	25 + 16 + 9 + 4 + 1 = 55.
2	D	(3 × 4 / 2) × (4 × 5 / 2) = 6 × 10 = 60.
3	A	12 squares of size 1, 6 of size 2 and 2 of size 3: 12 + 6 + 2 = 20.
4	C	Total squares = 30; of these 16 are 1 × 1. So 30 − 16 = 14 (9 + 4 + 1).
5	C	The base is divided into 5 segments; choosing a start and an end gives C(6,2) = 15 triangles.
6	B	Each level gives (2 + 1)(2 + 2)/2 = 6 triangles from the apex; there are 3 levels: 6 × 3 = 18.

CHAPTER 6

Quick Revision

One page for the last day

Figure	Count
Squares in n × n grid	n(n+1)(2n+1)/6
Rectangles in m × n grid	m(m+1)n(n+1)/4
Triangles with p lines from the apex	(p+1)(p+2)/2
...plus q lines parallel to the base	× (q + 1)
Tilted squares	Count diamonds formed by diagonals separately

PYQ RULE: English 2022 Q32: 4 × 4 grid with diagonals: 30 axis-aligned squares + 5 tilted squares = 35.

ROUTINE: Exam-day routine: count by size; add tilted squares; check with the formula. 45 seconds per question.